

TERTHO GHOSH

Computer Science & Engineering Graduate | Cybersecurity | IT & Networking | Web Development

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PROFESSIONAL SUMMARY

Computer Science and Engineering graduate from Jagannath University with a 3.88/4.00 CGPA and hands-on exposure to IT systems, databases, operating systems, networking, cybersecurity, and software development. Completed cybersecurity training as a Research Trainee at the ITRRC Cybersecurity Research Lab, gaining practical familiarity with Linux, security tools, OSINT, network reconnaissance, access control, and security concepts. Independently designed, developed, tested, and deployed multiple technical systems, including a cloud-native microservices-based plagiarism and AI-content detection platform. Strong analytical, problem-solving, documentation, and team coordination skills, with leadership experience managing a 6-member finance team and financial operations for an event involving 500+ people. Seeking to begin a career in IT Audit, General IT Controls (GITC), IT Risk, IT Controls, and Information Security.

EDUCATION

Bachelor of Science in Computer Science & Engineering | Jagannath University, Dhaka | 2026

CGPA: 3.88/4.00 | 7th Semester Completed | 8th Semester Examination Completed | Final Result Expected: December 2026

Higher Secondary Certificate | Notre Dame College, Dhaka | 2020

GPA: 5.00/5.00

Secondary School Certificate | Lalmohan Secondary School, Lalmohan, Bhola | 2018

GPA: 5.00/5.00

TECHNICAL SKILLS

IT & Operating Systems: Ubuntu, Kali Linux, Linux Command Line, Operating System Fundamentals, System Troubleshooting, Server Fundamentals

Networking & Security: TCP/IP, OSI Model, IPv4/IPv6, Subnetting, OSPF, BGP, Network Security, Cryptography, Authentication, Access Control, OSINT, Network Reconnaissance

Database: SQL, PostgreSQL, MySQL, Database Management, Database Design, Data Validation

Programming: Competitive programming, problem solving over 500 in Codeforces, Codechef, Leetcode and other platforms.
Max: Pupil in codeforces

Development & Infrastructure: Docker, Microservices, FastAPI, React, Git, GitHub, Supabase, XAMPP, Cisco Packet Tracer

Data & Machine Learning: Pandas, NumPy, Scikit-learn, NLP, Machine Learning, Data Analysis, Torch

IT Audit Areas of Interest: IT Audit, General IT Controls (GITC), IT Controls, IT Risk, Access Controls, Change Management, IT Operations, Audit Evidence, Control Testing

RESEARCH & CYBERSECURITY EXPERIENCE

Research Trainee | ITRRC Cybersecurity Research Lab, Jagannath University | January 2025 – July 2025

- Joined the founding cohort of the university's cybersecurity research lab and completed approximately 6 months of hands-on cybersecurity training.
- Developed practical familiarity with Ubuntu and Kali Linux, including Linux commands, user management, privileges, and system administration fundamentals.
- Gained hands-on exposure to cybersecurity tools and techniques involving OSINT, reconnaissance, and port sweeping.
- Studied active and passive attacks, their characteristics, objectives, and security implications.
- Practiced Linux-based access control and privilege management, including users, permissions, and sudo.
- Developed foundational understanding of operating-system security, network security, reconnaissance, and security assessment concepts.

LEADERSHIP EXPERIENCE

Head — Budget & Finance Management Team | TEDxJnU | 2025

- Led a 6-member finance and budget team supporting an event involving 50+ organizers, volunteers, and participants.
- Prepared the overall project budget estimation, identifying expected costs and financial requirements across multiple teams.
- Managed registration fees and sponsorship funds and allocated finances to teams according to operational requirements.
- Maintained records of project expenses and monitored financial transactions against the overall budget.
- Prepared the final financial report covering the event's financial activities and expenditures.

- Managed the event-day registration desk and coordinated food distribution and related transactions for participants and volunteers.

Treasurer | IEEE Student Branch, Jagannath University | January 2025 – February 2026

- Supported treasury and financial coordination activities for the student branch.
- Maintained and organised financial information related to branch activities.
- Coordinated with team members regarding event-related financial requirements and administrative activities.

SELECTED PROJECTS

PlagioScale — Plagiarism & AI-Generated Content Detection Platform | |

Python | FastAPI | React | PostgreSQL | Redis | Docker | Microservices | Prometheus | Grafana

- Planned, designed, developed, tested, and documented a cloud-native plagiarism and AI-generated-content detection platform.
- Designed a microservices architecture consisting of API, background worker, autoscaling, monitoring, and frontend services, with PostgreSQL and Redis supporting the system.
- Implemented a hybrid plagiarism-detection pipeline combining TF-IDF lexical similarity and SBERT semantic similarity, alongside AI-content detection using transformer-based models and stylometric features.
- Implemented security controls including JWT authentication, CSRF protection, role-based authorization, rate limiting, file validation, session invalidation, and audit logging.
- Built automated testing across backend and frontend components.
- Implemented monitoring using Prometheus and Grafana and created an administrative audit-log interface for tracking system activity.

CVInsight — AI Resume Analysis & Job Matching Platform | cvinsight-io.streamlit.app |

Python | JSON | NLP | Machine Learning | Streamlit

- Independently planned and developed the complete project from initial concept and system design through implementation, testing, deployment, and hosting.
- Built an automated resume analysis and job-matching system using NLP, machine learning, structured data processing, and feature engineering.
- Processed 6,241 human-labelled resume–job-description pairs for model development and evaluation.
- Developed 384-dimensional embeddings and incorporated 12 engineered features into the matching pipeline.
- Built an XGBoost-based scoring system producing a 0–100 compatibility score for resume–job matching.
- Deployed the completed application as a publicly accessible Streamlit application.

TrueMedic — Doctor Verification System | Android App |

Supabase | SQL | API | Authentication

- Developed a database-driven application for verifying doctor information using API-based data retrieval.
- Designed and managed structured database records for application data.
- Implemented authentication and access-control functionality.
- Applied data validation and structured workflows to improve the reliability of information processing.

Network Routing & Security Laboratory | Lab Work |

Cisco Packet Tracer | IPv4/IPv6 | OSPF | BGP

- Designed and configured multi-router IPv4/IPv6 network topologies using Cisco Packet Tracer.
- Configured and analysed RIP, OSPF and BGP routing protocols.
- Performed IP addressing, subnetting, routing-table analysis, and connectivity troubleshooting.

RELEVANT COURSEWORK

Database Management Systems | Operating Systems | Computer Networks | Network Security | Cryptography | Software Engineering | Data Mining & Data Warehousing | Cloud Computing | Internet of Things | Computer Architecture | Network Switching and Routing

CERTIFICATIONS & TECHNICAL ACTIVITIES

- Cisco Introduction to Cybersecurity — Cisco Networking Academy
- ICPC 2024 — DIU: Participant
- IUPC 2025 — KUET: Participant
- IUPC 2025 — AUST: Participant
- Participated in “PTIB Civic Tech”, “Sustainovate” and other technology-focused programmes and events.